

Library Management System using SQL

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Project Overview

Project Title: Library Management System

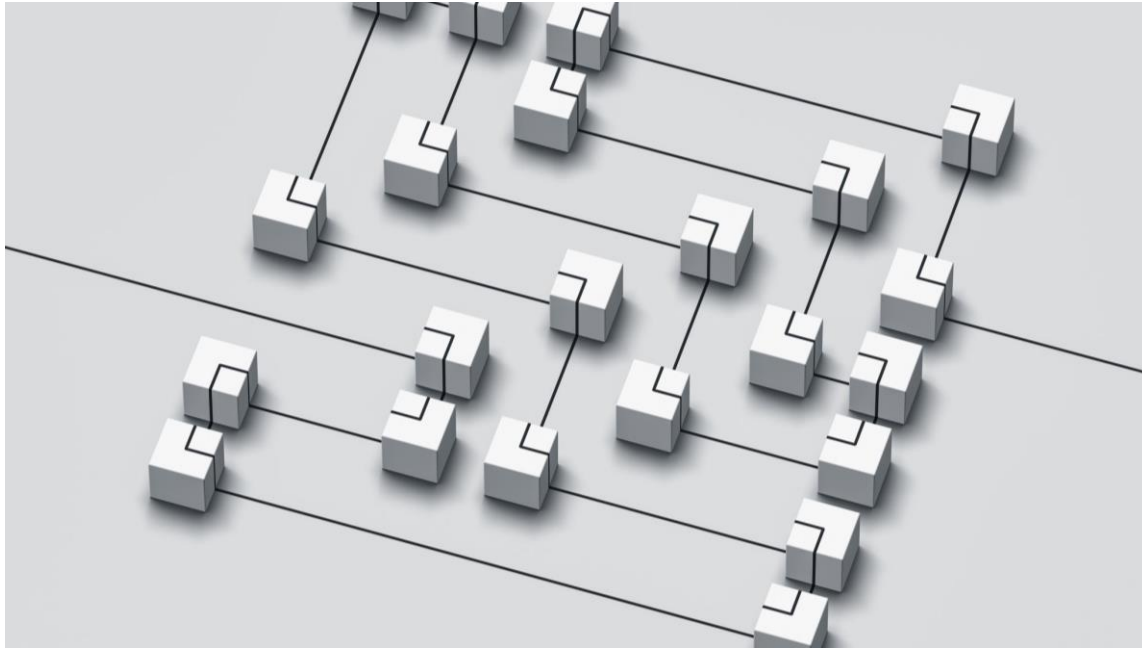
Objective:

- Implement a structured database for library management
- Perform CRUD operations
- Execute advanced SQL queries for analysis

Project Objectives

- **Database Setup** – Creating and populating tables
- **CRUD Operations** – Insert, Read, Update, Delete
- **CTAS (Create Table As Select)** – Generate new tables based on queries
- **Advanced SQL Queries** – Perform data analysis and reporting

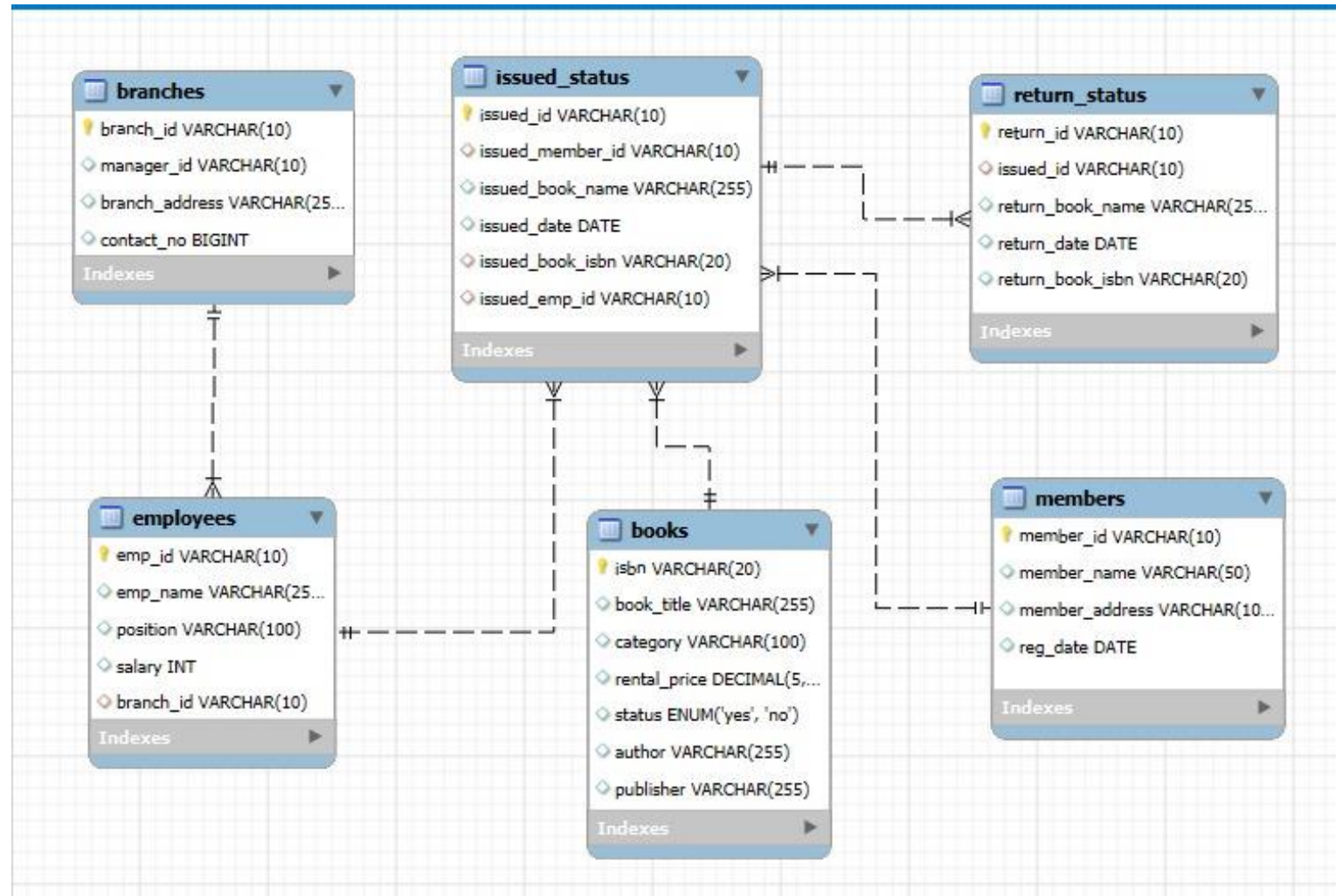
Database Schema & ERD



- **Tables Created:**
 - Branches
 - Employees
 - Members
 - Books
 - Issued Status
 - Return Status
- **Relationships & Foreign Keys**
 - Employees linked to Branches
 - Issued books linked to Members & Employees
 - Returned books linked to Issued Status

ERD

Diagram



Database Setup - SQL Commands

- Database Creation:

```
1 • CREATE DATABASE library_management_analysis
```

```
2
```

Creating books table

```
3 CREATE TABLE books (  
4     isbn VARCHAR(20) PRIMARY KEY,  
5     book_title VARCHAR(255),  
6     category VARCHAR(100),  
7     rental_price DECIMAL(5,2),  
8     status ENUM('yes', 'no'),  
9     author VARCHAR(255),  
10    publisher VARCHAR(255)  
11 );
```

Result Grid							
Filter Rows:							
Edit: Export/Import: Wrap Cell Content:							
	isbn	book_title	category	rental_price	status	author	publisher
▶	978-0-06-025492-6	Where the Wild Things Are	Children	3.50	yes	Maurice Sendak	HarperCollins
	978-0-06-112008-4	To Kill a Mockingbird	Classic	5.00	yes	Harper Lee	J.B. Lippincott & Co.
	978-0-06-112241-5	The Kite Runner	Fiction	5.50	yes	Khaled Hosseini	Riverhead Books
	978-0-06-440055-8	Charlotte's Web	Children	4.00	yes	E.B. White	Harper & Row
	978-0-09-957807-9	A Game of Thrones	Fantasy	7.50	yes	George R.R. Martin	Bantam
	978-0-14-027526-3	A Tale of Two Cities	Classic	4.50	yes	Charles Dickens	Penguin Books
	978-0-14-044930-3	The Histories	History	5.50	yes	Herodotus	Penguin Classics
	978-0-14-118776-1	One Hundred Years of Solitude	Literary Fiction	6.50	yes	Gabriel Garcia Marquez	Penguin Books
	978-0-14-143951-8	Pride and Prejudice	Classic	5.00	yes	Jane Austen	Penguin Classics
	978-0-141-44171-6	Jane Eyre	Classic	4.00	yes	Charlotte Bronte	Penguin Classics
	978-0-19-280551-1	The Guns of August	History	7.00	yes	Barbara W. Tuchman	Oxford University ...
	978-0-307-37840-1	The Alchemist	Fiction	2.50	yes	Paulo Coelho	HarperOne
	978-0-307-58837-1	Sapiens: A Brief History of Hu...	History	8.00	no	Yuval Noah Harari	Harper Perennial
	978-0-330-25864-8	Animal Farm	Classic	5.50	yes	George Orwell	Penguin Books
	978-0-345-39180-3	Dune	Science Fiction	8.50	yes	Frank Herbert	Ace
	978-0-375-41398-8	The Diary of a Young Girl	History	6.50	no	Anne Frank	Bantam
	978-0-375-50167-0	The Road	Dystopian	7.00	yes	Cormac McCarthy	Vintage
	978-0-385-33312-0	The Shining	Horror	6.00	yes	Stephen King	Doubleday

Creating branches table

```
13 • CREATE TABLE branches (  
14     branch_id VARCHAR(10) PRIMARY KEY,  
15     manager_id VARCHAR(10),  
16     branch_address VARCHAR(255),  
17     contact_no BIGINT  
18 );
```

Result Grid   Filter Rows: Edit: 				
	branch_id	manager_id	branch_address	contact_no
▶	B001	E109	123 Main St	919099988676
	B002	E109	456 Elm St	919099988677
	B003	E109	789 Oak St	919099988678
	B004	E110	567 Pine St	919099988679
	B005	E110	890 Maple St	919099988680
⊙	NULL	NULL	NULL	NULL

Creating employees table

```
20 • CREATE TABLE employees (  
21     emp_id VARCHAR(10) PRIMARY KEY,  
22     emp_name VARCHAR(255),  
23     position VARCHAR(100),  
24     salary INT,  
25     branch_id VARCHAR(10),  
26     FOREIGN KEY (branch_id) REFERENCES branches(branch_id)  
27 );
```

Result Grid					
Filter Rows:					
	emp_id	emp_name	position	salary	branch_id
▶	E101	John Doe	Clerk	60000	B001
	E102	Jane Smith	Clerk	45000	B002
	E103	Mike Johnson	Librarian	55000	B001
	E104	Emily Davis	Assistant	40000	B001
	E105	Sarah Brown	Assistant	42000	B001
	E106	Michelle Ramirez	Assistant	43000	B001
	E107	Michael Thompson	Clerk	62000	B005
	E108	Jessica Taylor	Clerk	46000	B004
	E109	Daniel Anderson	Manager	57000	B003
	E110	Laura Martinez	Manager	41000	B005
	E111	Christopher Lee	Assistant	65000	B005
⊙	NULL	NULL	NULL	NULL	NULL

Creating issued_status table

```
29 • CREATE TABLE issued_status (  
30     issued_id VARCHAR(10) PRIMARY KEY,  
31     issued_member_id VARCHAR(10),  
32     issued_book_name VARCHAR(255),  
33     issued_date DATE,  
34     issued_book_isbn VARCHAR(20),  
35     issued_emp_id VARCHAR(10),  
36     FOREIGN KEY (issued_member_id) REFERENCES members(member_id),  
37     FOREIGN KEY (issued_book_isbn) REFERENCES books(isbn),  
38     FOREIGN KEY (issued_emp_id) REFERENCES employees(emp_id)  
39 );
```

Result Grid						
Filter Rows:						
Edit: Export/Import: Wrap Cell Content:						
	issued_id	issued_member_id	issued_book_name	issued_date	issued_book_isbn	issued_emp_id
▶	IS106	C106	Animal Farm	2024-03-10	978-0-330-25864-8	E104
	IS107	C107	One Hundred Years of Solitude	2024-03-11	978-0-14-118776-1	E104
	IS108	C108	The Great Gatsby	2024-03-12	978-0-525-47535-5	E104
	IS109	C109	Jane Eyre	2024-03-13	978-0-141-44171-6	E105
	IS110	C110	The Alchemist	2024-03-14	978-0-307-37840-1	E105
	IS111	C109	Harry Potter and the Sorcerers Stone	2024-03-15	978-0-679-76489-8	E105
	IS112	C109	A Game of Thrones	2024-03-16	978-0-09-957807-9	E106
	IS113	C109	A Peoples History of the United States	2024-03-17	978-0-393-05081-8	E106
	IS114	C109	The Guns of August	2024-03-18	978-0-19-280551-1	E106
	IS115	C109	The Histories	2024-03-19	978-0-14-044930-3	E107
	IS116	C110	Guns, Germs, and Steel: The Fates of...	2024-03-20	978-0-393-91257-8	E107
	IS117	C110	1984	2024-03-21	978-0-679-64115-3	E107
	IS118	C101	Pride and Prejudice	2024-03-22	978-0-14-143951-8	E108
	IS119	C110	Brave New World	2024-03-23	978-0-452-28240-7	E108
	IS120	C110	The Road	2024-03-24	978-0-670-81302-4	E108
	IS122	C102	Fahrenheit 451	2024-03-26	978-0-451-52993-5	E109
	IS123	C103	Dune	2024-03-27	978-0-345-39180-3	E109
	IS124	C104	Where the Wild Things Are	2024-03-28	978-0-06-025492-6	E110

Creating members table

```
41 • CREATE TABLE members (  
42     member_id VARCHAR(10) PRIMARY KEY,  
43     member_name VARCHAR(50),  
44     member_address VARCHAR(100),  
45     reg_date DATE  
46 );
```

Result Grid				
Filter Rows:				
Edit:				
Export/Import:				
	member_id	member_name	member_address	reg_date
▶	C101	Alice Johnson	Fatima Jinnah Women University Rawalpindi	2021-05-15
	C102	Bob Smith	456 Elm St	2021-06-20
	C103	Carol Davis	789 Oak St	2021-07-10
	C104	Dave Wilson	567 Pine St	2021-08-05
	C105	Eve Brown	890 Maple St	2021-09-25
	C106	Frank Thomas	234 Cedar St	2021-10-15
	C107	Grace Taylor	345 Walnut St	2021-11-20
	C108	Henry Anderson	456 Birch St	2021-12-10
	C109	Ivy Martinez	567 Oak St	2022-01-05
	C110	Jack Wilson	678 Pine St	2022-02-25
	C118	Sam	133 Pine St	2024-06-01
	C119	John	143 Main St	2024-05-01
	C120	sam	145 Main St	2024-06-01
	C121	john	133 Main St	2024-05-01
	NULL	NULL	NULL	NULL

Creating return_status table

```
49 • CREATE TABLE return_status (  
50     return_id VARCHAR(10) PRIMARY KEY,  
51     issued_id VARCHAR(10),  
52     return_book_name VARCHAR(255),  
53     return_date DATE,  
54     return_book_isbn VARCHAR(20),  
55     FOREIGN KEY (issued_id) REFERENCES issued_status(issued_id)  
56 );
```

Result Grid					
Filter Rows:					
Edit:    					
	return_id	issued_id	return_book_name	return_date	return_book_isbn
▶	RS104	IS106	NULL	2024-05-01	NULL
	RS105	IS107	NULL	2024-05-03	NULL
	RS106	IS108	NULL	2024-05-05	NULL
	RS107	IS109	NULL	2024-05-07	NULL
	RS108	IS110	NULL	2024-05-09	NULL
	RS109	IS111	NULL	2024-05-11	NULL
	RS110	IS112	NULL	2024-05-13	NULL
	RS111	IS113	NULL	2024-05-15	NULL
	RS112	IS114	NULL	2024-05-17	NULL
	RS113	IS115	NULL	2024-05-19	NULL
	RS114	IS116	NULL	2024-05-21	NULL
	RS115	IS117	NULL	2024-05-23	NULL
	RS116	IS118	NULL	2024-05-25	NULL
	RS117	IS119	NULL	2024-05-27	NULL
	RS118	IS120	NULL	2024-05-29	NULL
•	NULL	NULL	NULL	NULL	NULL

Project Task

2. CRUD Operations

- Task 1. Create a New Book Record-- "978-1-60129-456-2', 'To Kill a Mockingbird', 'Classic', 6.00, 'yes', 'Harper Lee', 'J.B. Lippincott & Co.')"
- Task 2: Update an Existing Member's Address
- Task 3: Delete a Record from the Issued Status Table-- Objective: Delete the record with issued_id = 'IS104' from the issued_status table
- Task 4: Retrieve All Books Issued by a Specific Employee-- Objective: Select all books issued by the employee with emp_id = 'E101'.
- Task 5: List Members Who Have Issued More Than One Book-- Objective: Use GROUP BY to find members who have issued more than one book.

3. CTAS (Create Table As Select)

- Task 6: Create Summary Tables**: Used CTAS to generate new tables based on query results - each book and total book_issued_cnt

4. Data Analysis & Findings

- Task 7. **Retrieve All Books in a Specific Category:
- Task 8: Find Total Rental Income by Category:
- Task 9. **List Members Who Registered in the Last 180 Days**:
- Task 10: List Employees with Their Branch Manager's Name and their branch details
- Task 11. Create a Table of Books with Rental Price Above a Certain Threshold
- Task 12: Retrieve the List of Books Not Yet Returned

2. CRUD Operations

CRUD Operations

1. CREATE (Insert Data)

- Adds new records to a table.
- Used to store new books, users, or transactions in the database.

2. READ (Retrieve Data)

- Fetches data from the database without modifying it.
- Used to search for books, check availability, or view user details.

3. UPDATE (Modify Data)

- Changes existing records in a table.
- Used to update book availability, modify user details, or edit rental prices.

4. DELETE (Remove Data)

- Deletes records from a table permanently.
- Used to remove outdated books, delete user accounts, or erase old transactions.
- Would you like this formatted into PowerPoint slides? 😊

Task 1:

Create a New Book Record-- "978-1-60129-456-2', 'To Kill a Mockingbird', 'Classic', 6.00, 'yes', 'Harper Lee', 'J.B. Lippincott & Co.')

```
17  insert into
18  books(isbn,    book_title, category,    rental_price,    status, author, publisher)
19  values
20  ('978-1-60129-456-2', 'To Kill a Mockingbird', 'Classic', 6.00, 'yes', 'Harper Lee', 'J.B. Lippincott & Co.');
```

Task 1:

New record is added in books table:

```
21 ● SELECT
22      *
23 FROM
24      books
```

Result Grid							
Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:	
isbn	book_title	category	rental_price	status	author	publisher	
978-0-452-28240-7	Brave New World	Dystopian	6.50	yes	Aldous Huxley	Harper Perennial	
978-0-525-47535-5	The Great Gatsby	Classic	8.00	yes	F. Scott Fitzgerald	Scribner	
978-0-553-29698-2	The Catcher in the Rye	Classic	7.00	yes	J.D. Salinger	Little, Brown and ...	
978-0-553-57340-1	1984	Dystopian	6.50	yes	George Orwell	Penguin Books	
978-0-670-81302-4	The Road	Dystopian	7.00	yes	Cormac McCarthy	Knopf	
978-0-679-64115-3	1984	Dystopian	6.50	yes	George Orwell	Penguin Books	
978-0-679-76489-8	Harry Potter and the Sorcerer...	Fantasy	7.00	yes	J.K. Rowling	Scholastic	
978-0-679-77644-3	Beloved	Fiction	6.50	yes	Toni Morrison	Knopf	
978-0-7432-4722-4	The Da Vinci Code	Mystery	8.00	yes	Dan Brown	Doubleday	
978-0-7432-4722-5	Angels & Demons	Mystery	7.50	yes	Dan Brown	Doubleday	
978-0-7432-7356-4	The Hobbit	Fantasy	7.00	yes	J.R.R. Tolkien	Houghton Mifflin H...	
978-0-7432-7357-1	1491: New Revelations of the...	History	6.50	no	Charles C. Mann	Vintage Books	
978-0-7434-7679-3	The Stand	Horror	7.00	yes	Stephen King	Doubleday	
978-1-60129-456-2	To Kill a Mockingbird	Classic	6.00	yes	Harper Lee	J.B. Lippincott & Co.	
NULL	NULL	NULL	NULL	NULL	NULL	NULL	

Task 2:

Update an Existing Member's Address:

```
28 UPDATE members
29 SET
30     member_address = 'Fatima Jinnah Women Uniersity Rawalpindi'
31 WHERE
32     member_id = 'C101'
```

Result Grid	Filter Rows:	Edit:	Export/Import:
member_id	member_name	member_address	reg_date
C101	Alice Johnson	Fatima Jinnah Women Uniersity Rawalpindi	2021-05-15
C102	Bob Smith	456 Elm St	2021-06-20
C103	Carol Davis	789 Oak St	2021-07-10
C104	Dave Wilson	567 Pine St	2021-08-05
C105	Eve Brown	890 Maple St	2021-07-10
C106	Frank Thomas	234 Cedar St	2021-10-15
C107	Grace Taylor	345 Walnut St	2021-11-20
C108	Henry Anderson	456 Birch St	2021-12-10
C109	Ivy Martinez	567 Oak St	2022-01-05
C110	Jack Wilson	678 Pine St	2022-02-25
C118	Sam	133 Pine St	2024-06-01
C119	John	143 Main St	2024-05-01
C120	sam	145 Main St	2024-06-01
C121	john	133 Main St	2024-05-01
NULL	NULL	NULL	NULL

Task 3:

Delete a Record from the Issued Status Table:

```
DELETE FROM issued_status  
WHERE  
    issued_id = 'IS121'
```

Task 4:

Retrieve All Books Issued by a Specific Employee

```
46  SELECT
47      *
48  FROM
49      issued_status
50  WHERE
51      issued_emp_id = 'E101'
```

Result Grid						
	issued_id	issued_member_id	issued_book_name	issued_date	issued_book_isbn	issued_emp_id
▶	IS130	C106	Moby Dick	2024-04-03	978-0-451-52994-2	E101
	IS131	C106	To Kill a Mockingbird	2024-04-04	978-0-06-112008-4	E101
*	NULL	NULL	NULL	NULL	NULL	NULL

Task 5:

List Members Who Have Issued More Than One Book:

```
56 SELECT
57     issued_emp_id, COUNT(issued_id) AS total_books_issued
58 FROM
59     issued_status
60 GROUP BY issued_emp_id
61 HAVING COUNT(issued_id) > 1
```

Result Grid			Filter Rows:
	issued_emp_id	total_books_issued	
▶	E101	2	
	E102	2	
	E104	4	
	E105	4	
	E106	6	
	E107	3	
	E108	4	
	E109	2	
	E110	6	

3. CTAS (Create Table As Select)

Task 6: Create Summary Tables : Used CTAS to generate new tables based on query results - each book and total book_issued_cnt

```
68 CREATE TABLE books_count AS SELECT b.isbn, b.book_title, COUNT(ist.issued_id) AS no_issued FROM
69     books AS b
70     JOIN
71     issued_status AS ist ON ist.issued_book_isbn = b.isbn
72 GROUP BY 1 , 2
```

```
75 SELECT
```

```
76     *
```

```
77 FROM
```

```
78     books_count
```

Result Grid				Filter Rows:	Export:
	isbn	book_title	no_issued		
▶	978-0-06-025492-6	Where the Wild Things Are	1		
	978-0-06-112008-4	To Kill a Mockingbird	1		
	978-0-06-112241-5	The Kite Runner	1		
	978-0-06-440055-8	Charlotte's Web	1		
	978-0-09-957807-9	A Game of Thrones	1		
	978-0-14-027526-3	A Tale of Two Cities	1		
	978-0-14-044930-3	The Histories	1		
	978-0-14-118776-1	One Hundred Years of Solitude	1		
	978-0-14-143951-8	Pride and Prejudice	1		
	978-0-141-44171-6	Jane Eyre	1		
	978-0-19-280551-1	The Guns of August	1		
	978-0-307-37840-1	The Alchemist	1		
	978-0-307-58837-1	Sapiens: A Brief History of Hu...	1		
	978-0-330-25864-8	Animal Farm	2		
	978-0-345-39180-3	Dune	1		
	978-0-375-41398-8	The Diary of a Young Girl	1		
	978-0-393-05081-8	A Peoples History of the Unite...	1		

Task 8:

Find Total Rental Income by Category:

```
93  SELECT
94      b.category, SUM(b.rental_price) AS rental_price, COUNT(*)
95  FROM
96      books AS b
97      JOIN
98      issued_status AS ist ON ist.issued_book_isbn = b.isbn
99  GROUP BY 1
```

Result Grid			
	category	rental_price	COUNT(*)
▶	Children	7.50	2
	Classic	59.00	10
	Fiction	14.50	3
	Fantasy	28.50	4
	History	49.50	7
	Literary Fiction	6.50	1
	Science Fiction	8.50	1
	Dystopian	25.50	4
	Mystery	7.50	1
	Horror	7.00	1

Task 9:



List Members Who Registered in the Last 180 Days:

```
104  SELECT * FROM members
105  WHERE reg_date >= CURRENT_DATE - INTERVAL '180 days'
106
107  INSERT INTO members(member_id, member_name, member_address, reg_date)
108  VALUES
109  ('C120', 'sam', '145 Main St', '2024-06-01'),
110  ('C121', 'john', '133 Main St', '2024-05-01');
111
```


Task 10:

List Employees with Their Branch Manager's Name and their branch details**:

```
SELECT
    e1.*, b.manager_id, e2.emp_name as manager
FROM
    employees AS e1
    JOIN
    branches AS b ON b.branch_id = e1.branch_id
join employees as e2
```

Result Grid							
Filter Rows:							
Export:  Wrap Cell Content: 							
	emp_id	emp_name	position	salary	branch_id	manager_id	manager
▶	E111	Christopher Lee	Assistant	65000	B005	E110	John Doe
	E110	Laura Martinez	Manager	41000	B005	E110	John Doe
	E107	Michael Thompson	Clerk	62000	B005	E110	John Doe
	E108	Jessica Taylor	Clerk	46000	B004	E110	John Doe
	E109	Daniel Anderson	Manager	57000	B003	E109	John Doe
	E102	Jane Smith	Clerk	45000	B002	E109	John Doe
	E106	Michelle Ramirez	Assistant	43000	B001	E109	John Doe
	E105	Sarah Brown	Assistant	42000	B001	E109	John Doe
	E104	Emily Davis	Assistant	40000	B001	E109	John Doe
	E103	Mike Johnson	Librarian	55000	B001	E109	John Doe
	E101	John Doe	Clerk	60000	B001	E109	John Doe
	E111	Christopher Lee	Assistant	65000	B005	E110	Jane Smith
	E110	Laura Martinez	Manager	41000	B005	E110	Jane Smith
	E107	Michael Thompson	Clerk	62000	B005	E110	Jane Smith
	E108	Jessica Taylor	Clerk	46000	B004	E110	Jane Smith

Task 11:

Create a Table of Books with Rental Price Above a Certain Threshold

```
CREATE TABLE books_price_greater_than_seven AS SELECT * FROM
books
WHERE
rental_price > 7
```

```
SELECT
*
FROM
books_price_greater_than_seven
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	isbn	book_title	category	rental_price	status	author	publisher
▶	978-0-09-957807-9	A Game of Thrones	Fantasy	7.50	yes	George R.R. Martin	Bantam
	978-0-307-58837-1	Sapiens: A Brief History of Humankind	History	8.00	no	Yuval Noah Harari	Harper Perennial
	978-0-345-39180-3	Dune	Science Fiction	8.50	yes	Frank Herbert	Ace
	978-0-393-05081-8	A Peoples History of the United States	History	9.00	yes	Howard Zinn	Harper Perennial
	978-0-525-47535-5	The Great Gatsby	Classic	8.00	yes	F. Scott Fitzgerald	Scribner
	978-0-7432-4722-4	The Da Vinci Code	Mystery	8.00	yes	Dan Brown	Doubleday
	978-0-7432-4722-5	Angels & Demons	Mystery	7.50	yes	Dan Brown	Doubleday

Task 12:

Retrieve the List of Books Not Yet Returned

```
136  SELECT
137      distinct ist.issued_book_name
138  FROM
139      issued_status AS ist
140      left JOIN
141      return_status AS rs ON ist.issued_id = rs.issued_id
142  where rs.return_id is null
```

Result Grid		Filter Rows
	issued_book_name	
▶	Fahrenheit 451	
	Dune	
	Where the Wild Things Are	
	The Kite Runner	
	Charlotte's Web	
	Beloved	
	A Tale of Two Cities	
	The Stand	
	Moby Dick	
	To Kill a Mockingbird	
	The Hobbit	
	Angels & Demons	
	The Diary of a Young Girl	
	Sapiens: A Brief History o...	
	1491: New Revelations of...	
	The Catcher in the Rye	
	The Great Gatsby	